TECHNICAL REPORT

MGHRIBAI :

DISCOVER MOROCCO. YOUR WAY.

An AI-Assisted Tourism Platform for Morocco

HackApertus 2026 — Track 2B

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GitHub Repository: https://github.com/elhaddouchikarima-debug/MghribAI

Event: Online



1. Project Overview

1.1 Introduction

Morocco is a country rich in cultural heritage, diverse landscapes, traditional cuisine, and unique travel experiences. However, planning a trip that matches a traveler's interests, schedule, and budget can require information from multiple sources.

MghribAI aims to simplify this process through an AI-assisted tourism platform dedicated to discovering Morocco and exploring personalized travel experiences.

1.2 Problem Statement

Travelers may face difficulties finding relevant destinations, comparing experiences, organizing daily activities, and estimating the total cost of a trip. Information about attractions, accommodation, and local food can also be scattered across different platforms.

1.3 Proposed Solution

MghribAI combines a tourism-focused web interface with structured information about Moroccan destinations and an AI-assisted approach to travel planning.

The project aims to help users discover destinations, explore local experiences, and organize travel plans according to their preferences, trip duration, and budget.

1.4 Main Objectives

- Simplify the discovery of Moroccan destinations.
- Support personalized travel planning.
- Highlight Moroccan culture, cuisine, and local experiences.
- Organize activities and estimated travel expenses.
- Explore the use of open AI models for tourism assistance.

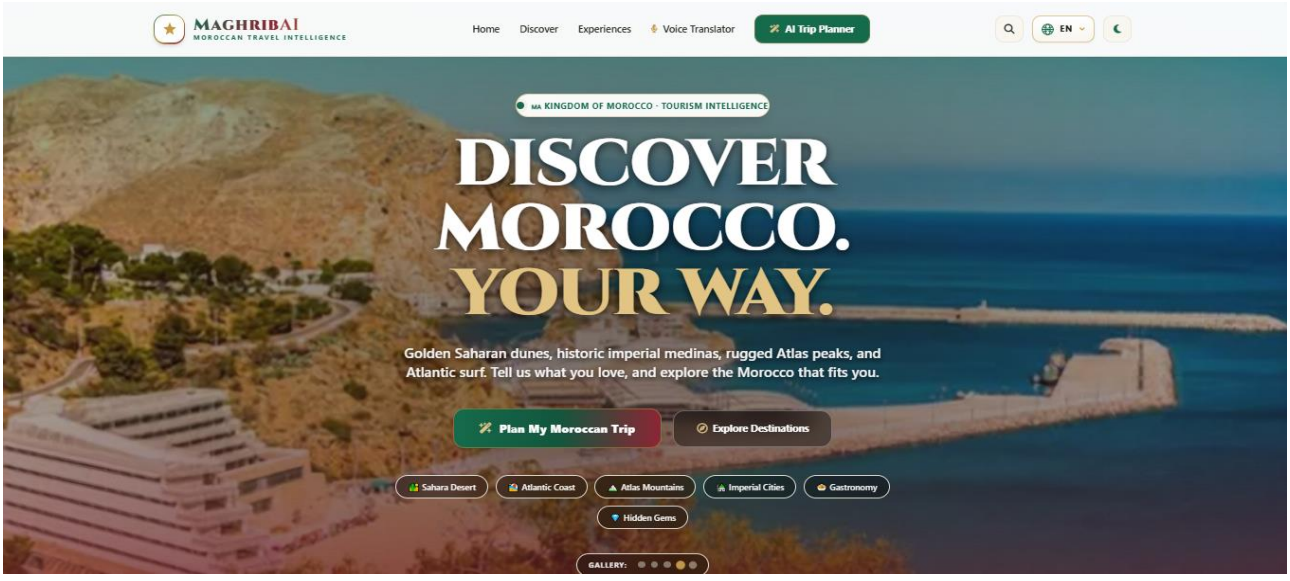


Figure 2. MghribAI homepage and user interface.

2. System Architecture

2.1 Architecture Overview

MghribAI is designed around a web interface, backend services, tourism data, and trip-planning logic. An AI integration layer is intended to support personalized recommendations when the model is configured and available.

The system receives user preferences, validates the submitted information, uses relevant tourism data, and prepares recommendations for presentation through the web interface.

2.2 Main Components

Frontend: Provides the interface through which travelers explore destinations and submit their preferences.

Backend: Handles application requests and coordinates the relevant services.

Preference Validation: Checks user inputs before they are processed.

Tourism Dataset: Organizes information about destinations and travel experiences.

Trip-Planning Logic: Uses available preferences and tourism information to support itinerary creation.

AI Integration: Intended to use Apertus for tourism assistance and itinerary generation, subject to the implemented configuration.

2.3 Data Flow

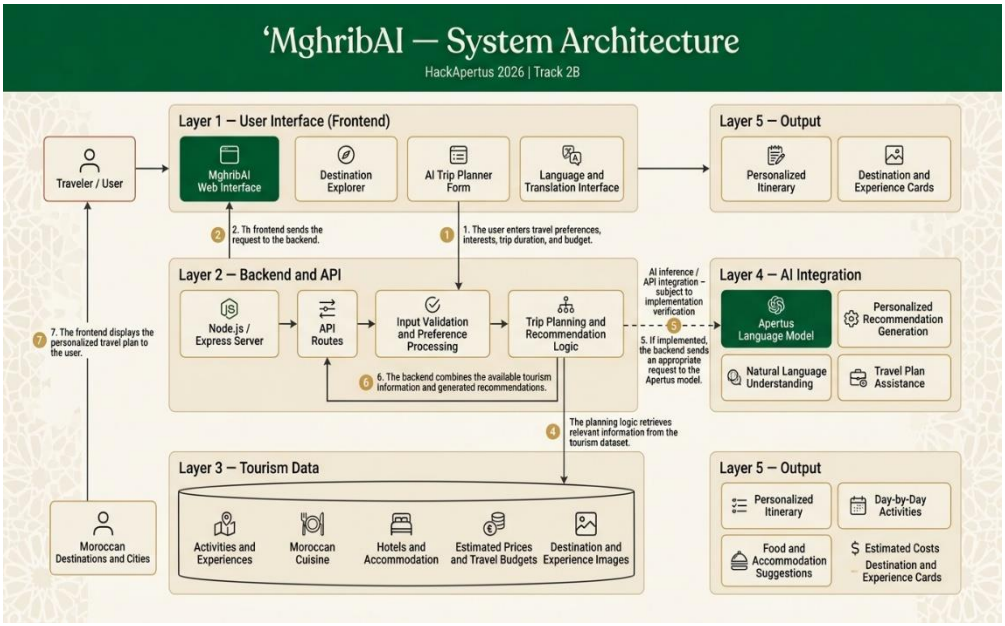


Figure 2. Conceptual architecture and information flow of MghribAI.

2.4 Target Deployment Architecture

The project targets **on-premise deployment**, where the application runs on infrastructure administered by its operating organization.

This deployment model could provide greater control over application execution and tourism data. However, full on-premise operation must be validated against the actual model-serving configuration and runtime dependencies.

Air-gapped operation and deployment on a sovereign Swiss cloud are not claimed as implemented.

3. Tourism Data and AI Integration

3.1 Tourism Data

Tourism information is central to the MghribAI experience. The project is designed to organize information that can help travelers discover Moroccan destinations and plan meaningful experiences.

The intended data categories include:

- **Destinations:** Moroccan cities, landmarks, and places of interest.
- **Activities:** Cultural visits, outdoor activities, and local experiences.
- **Cuisine:** Traditional Moroccan dishes and dining recommendations.
- **Accommodation:** Hotel and lodging suggestions.
- **Estimated Costs:** Potential expenses for activities, meals, and accommodation.
- **Images:** Photographs illustrating destinations and experiences.

The reliability of recommendations depends on the quality, completeness, and freshness of the underlying information. Data sources and image licenses should be documented before publication.

3.2 Role of Apertus

Apertus is an open multilingual language model developed by the Swiss AI Initiative. MghribAI explores its potential use in tourism assistance and personalized itinerary generation.

The intended workflow combines user preferences with relevant tourism information to produce clear and useful recommendations.

The model should be evaluated for factual reliability, adherence to user constraints, and the ability to distinguish verified information from estimates.

The exact model version, inference environment, and integration status must be documented according to the configuration actually tested.

Figure 4. Examples of destinations and travel experiences presented by MghribAI.



3.3 Data Reliability

Prices and accommodation information may change over time. Estimated costs should therefore be clearly identified, and the system should avoid presenting unverified availability or unsupported claims as facts.

4. Evaluation and Limitations

4.1 Evaluation Objectives

The quality of MghribAI should be evaluated through reproducible tests that measure the usefulness and reliability of its tourism recommendations.

The evaluation should consider the following criteria:

- **Budget Compliance:** Does the estimated itinerary cost respect the user's budget?
- **Constraint Satisfaction:** Does the plan follow the user's preferences and requested trip duration?
- **Data Grounding:** Are recommendations supported by the available tourism dataset?
- **Itinerary Completeness:** Does the plan include the requested information for each travel day?
- **Output Validity:** Does the generated response follow the expected structure?

4.2 Limitations

The project has several limitations that require attention:

- Tourism information and estimated prices may become outdated.
- AI-generated recommendations may contain inaccurate information.
- Recommendation quality depends on the tourism dataset.
- Model performance may vary across languages and requests.
- Local deployment depends on available hardware and software requirements.

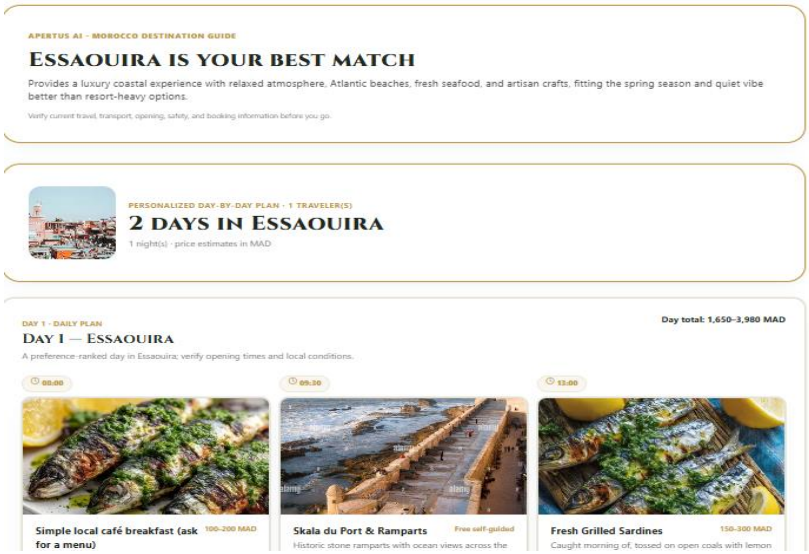


Figure 5. Example of the itinerary interface, if implemented.

5. Deployment, Future Work and References

Deployment and Reproducibility

MghribAI is designed as a web-based tourism assistant. Its deployment and AI integration depend on the final implementation and available infrastructure. The project aims to explore local or on-premise AI deployment in line with HackApertus Track 2B requirements.

Repository: https://github.com/elhaddouchikarima-debug/MghribAI

Future Work

- Improve tourism data accuracy and coverage.
- Enhance personalized, day-by-day itineraries.
- Complete and validate Apertus AI integration.
- Improve multilingual accessibility.
- Evaluate recommendation quality and budget consistency.

References

- MghribAI source code and project documentation.
- HackApertus 2026 — Track 2B.
- Official Apertus model documentation.
- Tourism information sources used in the dataset.